

The Hand That Gives

*An Indo-European etymology for the Greek-Celtic-Baltic hand cluster:
de-r- as a compositional body-part noun

Paper 1 of the *der-* project.

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Abstract

A recognized comparative cluster — Greek δῶρον "gift; palm-breadth," Hesychian δάρις "span of all fingers," Celtic **durno-* "fist, hand" (Old Irish *dorn*, Welsh *dwrn*, Breton *dorn/dourn*, Gaulish *Durnomagus*), and Latvian *dūre* "fist" — has been treated as a puzzle since Pokorny (IEW 203). Schrijver (1995: 58) lists these forms as "possible cognates" and describes the formal relation between δωρ- and δαρ- as "obscure." Matasóvič (2009, s.v. **durno-*) has concluded that no satisfactory Indo-European etymology exists and has classified the Celtic word as "probably a loanword from a non-IE language." This paper offers the missing etymology.

The body-part noun is the compositional **de-r-*: the bare hand-element **de-* of Paper 3 (independently attested as the directional particle **de/do* preserved in Greek -δε, Mycenaean *-de*, English *to*, Old Church Slavonic *do*) plus the locative/spatial extension **-r-* (supported by Lipp 2009, Bauhaus 2019, Kahl 2024). The body-part noun **de-r-* "at the place of the hand-direction" surfaces in the unmarked feminine singulative as Old Irish *derna*, Gaelic *deàrna* "palm of the hand," in the Greek zero-grade body-part form Hesychian δάρις "span of all fingers," and in Continental Celtic **DERNV[* (Bronze of Novallas, Stifter 2022). The Celtic compound stem **durn-* — attested in Insular Celtic **durno-* "fist" and Continental Gaulish *Durno-* — arises from regular Celtic morphology operating on the compositional stem; the *u* is morphological-stage Celtic, not inherited PIE root vocalism.

Greek δῶρον "gift" is the gift-noun **deh₃-ro-* derived from the verbal stem **deh₃-* (which Paper 3 decomposes as **de-* + verbal extension **-h₃-*). The body-part "palm-breadth" sense of δῶρον is preserved by family-association with the sibling body-part noun **de-r-*: two morphological siblings from the same bare element, coexisting in a speaker's lexicon and bleeding semantics across. Vitruvius's ancient causal chain (*De architectura* 2.3.3) is the synchronic Greek-internal evidence for that bleed. **der-* (nominal) and **deh₃-* (verbal) are siblings. The framework is offered for examination, not as a refutation of the standard view. We assess the probability of the Paper 1 thesis at 25–35%.

1. The question

A recognized comparative cluster has been sitting in the specialist literature for over a century without a satisfactory etymology. Pokorny's *Indogermanisches etymologisches Wörterbuch* (IEW 203) grouped together Greek δῶρον "length of the hand," Hesychian δάρις "span of all fingers," Old Irish *dorn*, Welsh *dwrn*, Gaulish *Durnomagus*, and Latvian *dūre* as members of a single comparative set. Schrijver (1995: 58), in the current authoritative monograph on British Celtic historical phonology, lists these forms as "possible cognates" and describes the formal relation between δωρ- and δαρ- as "obscure." Matasóvič (2009, s.v. **durno-*), the current standard reference on Proto-Celtic, states that the IE etymology "cannot be accepted" and classifies the Celtic word as "probably a loanword from a non-IE language." Two things are clear from this record: the comparative cluster is real, and the etymology is missing.

Three specific puzzles have driven the literature's difficulty. First, the relation between the Greek alternants. Greek δῶρον (with ω) and Hesychian δάρις (with α) are clearly related but the formal derivation connecting them has not been worked out to general satisfaction. Frisk (*GEW* I: 430–431) and the Spanish *Diccionario Griego-Español* both treat them as ablaut alternants, but no standard reconstruction has explained the alternation within the established laryngeal framework. Second, the Celtic *u*-vocalism. The consistent *u* of Celtic **durno-* and Continental Gaulish *Durno-* has resisted systematic derivation from either a bare **der-* or the give-verb **deh₃-*. Third, the semantic coherence. Why should the same root give "hand, fist, palm" in Celtic and Baltic but "gift" in Greek and Latin? The semantic gap has seemed too large for coincidence but too wide to bridge.

The present paper proposes a unified answer to all three puzzles under the bare-element architecture of the broader four-paper project. The formal relation between δωρ- and δαρ- is that of two sibling derivations from the same bare hand-element **de-* via different extensions: the verbal-derived gift-noun **deh₃-ro-* and the locative-derived body-part noun **de-r-*. The Celtic *u* is not inherited PIE root vocalism but the regular output of Celtic compound-stem morphology operating on the compositional stem **de-r-*. The semantic gap between "gift" and "hand" is not a gap but an inheritance: the giving and the open palm are two faces of the same physical act, derived from the same bare hand-element, and the coexistence of both senses in Greek literature preserves the family resemblance.

2. Historical context and the bare-element architecture

The framework does not introduce a new root. It recovers the comparative cluster Pokorny assembled and gives it the formal account that the nineteenth-century philologists, Pokorny himself, and Schrijver have all noted is missing.

Papers 2, 3, and 4 of this project have established the following. Paper 3 (Chavis & Claude 2026c) decomposes the give-verb **deh₃-* as bare **de-* (the directional particle attested as Greek allative *-δε*, Mycenaean *-de*, English *to*, Old Church Slavonic *do*) plus verbal extension **-h₃-*, and decomposes the contrastive suffix **-tero-* as the same bare **de-* plus locative **-r-* plus thematic **-o-*. Paper 2 (Chavis & Claude 2026b) demonstrates that **-tero-* is the systematic spatial-direction marker across the IE family. Paper 4 (Chavis & Claude 2026d) derives Hittite *dā-* "to take" from the directive member of an early-PIE paradigmatic pair on the same bare root: locative **de-r-* versus directive **de + a*, a structural type Bauhaus (2019: §5) has independently reconstructed for early PIE nominal roots.

Paper 1's contribution to this architecture is the lexical branch: the compositional **de-r-* (= **de-* + locative **-r-*) is the body-part noun "at the place of the hand-direction," and the comparative cluster Pokorny listed at IEW 203 is its surviving residue across six branches of the family. The framework does not require a new PIE root and no novel laryngeal phonology. It requires only the bare-element architecture itself: the same bare **de-* and the same locative **-r-* that Papers 2, 3, and 4 have independently motivated.

3. The Greek evidence

3.1 δῶπον: gift-noun with palm-breadth by family-association

Greek δῶπον is attested with two meanings from the earliest Homeric layer: "gift" (the standard word for a valued object exchanged or dedicated) and "palm-breadth" (a unit of measurement equal to four finger-widths, one-quarter of a foot), attested in *Iliad* 4.109, Hesiod's *Works and Days*, Nikander's *Theriaca* 398, Vitruvius's *De architectura* 2.3.3, and Pollux's *Onomasticon* 2.157. Pollux 2.144 records δῶπον additionally as an anatomical term for the front part of the palm, independently of the metrological tradition.

Under the framework's architecture, δῶπον "gift" is the regular post-laryngeal Greek outcome of **deh₃-ro-*: the o-grade verbal-stem **doh₃-ro-* undergoes regular laryngeal coloring and compensatory lengthening to yield **dō-ro-* > δῶπον. This is the standard derivation, unchanged. The body-part "palm-breadth" sense is preserved not by direct inheritance from a body-part root but by family-association: two sibling derivations from the same bare **de-* — the gift-noun **deh₃-ro-* and the locative body-part noun **de-r-* — coexisted in the Greek lexicon, shared phonological resemblance and a common semantic field (hand-action and hand-as-object), and bled semantics across in the way related words routinely do. Beekes's two-lemma solution — treating the gift-sense and the palm-breadth sense as etymologically unrelated — misses this family relationship.

Vitruvius supplies the most explicit ancient testimony to the perceived unity: "δῶρον Graeci appellant palmum, quod munus datio graece δῶρον appellatur, id autem semper geritur per manus palmam" (*De architectura* 2.3.3). His causal chain — the palm-breadth unit is called δῶρον because gift-giving is performed through the open palm — is the synchronic Greek-internal evidence for the semantic bleed having occurred. The directionality of his folk etymology (gift > palm-breadth) is the reverse of the historical derivation, but the perceived semantic unity is the relevant datum.

3.2 δάρις/δάρις: the primary Greek body-part witness

Hesychian δάρις/δάρις, glossed as "span of all fingers" (Attic equivalent σπθαμῆ), is the direct Greek-internal surface witness for the body-part noun **de-r-*. Under the framework, this is the *i*-stem derivative **de-r-i-s* with regular Greek treatment of the cluster: a zero-grade or *e*-grade formation on the compositional body-part noun, meaning "the span of the hand-direction." Where δῶρον preserves the verbal-derived gift-noun (with the long ω from laryngeal coloring of the verbal stem), δάρις preserves the body-part noun (with the α of the non-laryngeal formation). The formal relation between δωρ- and δαρ- that Schrijver called "obscure" is the family relation between two sibling derivations from **de-*: the verbal-stem gift-noun and the locative-stem body-part noun.

Two independent specialist sources recognize this as a coherent comparative set. Frisk (*GEW* I: 430–431) treats δάρις/δάρις as the *i*-stem derivative of δῶρον in its "palm-breadth" sense, with δαρ- as the ablaut alternant of δωρ-. The Spanish *Diccionario Griego-Español* makes the comparative argument explicit: the forms "hacen verosímil una alternancia." Two independent scholarly treatments, one working on Greek-internal evidence and one on Celtic comparative evidence, both recognize the cluster as a formal problem. The framework proposes the solution.

3.3 δάνος: the r/n-heteroclite n-stem alternant

Greek δάνος "gift, loan" provides Greek-internal evidence for an r/n-heteroclite morphology in the gift-noun family. Under the framework, the r-stem alternant is the strong form (**deh₃-r-*, giving δῶρον) and the n-stem alternant is the weak form (**deh₃-n-*, giving δάνος by regular zero-grade vocalization). Latin *dōnum* < **dōr-no-* represents the strong-stem n-formation. Stifter (2022: §8), in his etymological analysis of the Bronze of Novallas, registers a heteroclitic r/n-stem as a recognized morphological possibility for the **deh₃-* verbal family — Celtic-specialist published support for the morphological type the framework employs.

4. The Celtic evidence

4.1 The cognate set: u-vocalism is original in Celtic, but morphological

Celtic **durno-* "fist, hand" is attested with consistent *u*-vocalism in all four major branches: Old Irish *dorn*, Welsh *dwrn*, Breton *dorn/dourn*, Cornish *dorn*, and — critically for the chronology — Continental Celtic Gaulish *Durnomagus*, *Durnacos*, *Dagodurnus* in 1st-century-BCE inscriptions and classical authors. The Gaulish evidence is decisive: the *u* was already present in Continental Celtic before any Brythonic-internal sound change could have applied. As Schrijver (1995: 58) notes, this rules out the Morris Jones (1913: 87) raising rule **o > *u / __rn* as the source of the Brythonic vocalism.

Earlier versions of this paper attempted to derive the Celtic *u* from PIE root vocalism, trying both a bare **der-* with Morris Jones's raising rule and a laryngeal-extended **deuH-r-* with inherited *u*. Both positions were in tension with the bare-element architecture of Papers 2, 3, and 4, which reconstruct the body-part noun as compositional **de-* + locative **-r-* — a formation with no internal *u* or laryngeal. The present paper resolves the tension by accepting the bare-element segmentation as primary and explaining the Celtic *u* at the Celtic morphological level rather than at the PIE root level.

4.2 The framework's reconstruction: **de-r-* with morphological **u* in the compound stem

The Celtic forms reflect two formations on the same compositional stem **de-r-*. The *unmarked feminine singulative* **de-r-nā-* yields Old Irish *derna*, Gaelic *deàrna* "palm of the hand," and Continental Celtic **DERNV[* (Bronze of Novallas, Stifter 2022 §10; see §4.4 below). The *morphologically-marked compound stem* **der-u-* + singulative **-no-* yields **deruno-* > (with vowel reduction at the morpheme boundary) **durno-* "fist," lexicalizing the closed or grasping configuration of the hand. The Continental Gaulish onomastic forms *Durnomagus*, *Durnacos*, *Dagodurnus* preserve the compound stem **durn-* directly.

The framework does not commit to a single Celtic-internal mechanism for the *u* in the compound stem. Three live candidates exist. (a) The Celtic compound-first-element linking vowel: Celtic compounds show first elements in oblique stem-forms with various linking vowels documented across the family. (b) A derivational **-u-* extension on the nominal stem, parallel to standard PIE *u*-stem nominal formations. (c) The dative-locative **-u* preserved in lexicalized compound morphology: PIE **-ōi* > Common Celtic **-ūi* > apocopated **-u* (established for Insular Celtic and Celtiberian; Koch 2010 extends the chronology to the 8th–7th century BCE if Tartessian is Celtic). Any of these suffices for the framework. The relevant claim is the negative one: the Celtic *u* is morphological-stage Celtic, not inherited PIE root vocalism.

4.3 The compounds confirm "hand," not "fist"

The body-part meaning of the Celtic noun is confirmed beyond doubt by compound formations whose semantics admit no other reading. Proto-Celtic **are-durnon* (Old

Irish *aurdorn*, Welsh *arddwrn*, Breton *arzorn*) means "wrist" — literally "the part of the limb proximal to the hand." Proto-Celtic **ambi-durnos* (Old Irish *imdorn*) means "sword hilt" — literally "the around-the-hand thing." Neither compound makes sense with "fist" as the base element. A wrist is not "the part proximal to the fist"; a sword hilt is not "the part around the fist." Both are transparent and natural with "hand." The fist-meaning of the simplex **durno-* is one specialization of the hand-meaning, selected by the morphological compound-stem-form; the compounds preserve the unmarked "hand" sense because the compound boundary preempts the inner specialization.

Matasóvič's *Etymological Dictionary of Proto-Celtic* (2009, s.v. **durno-*) marks the form as of unknown etymology. The present hypothesis offers the missing Indo-European etymology. The Greek-Celtic comparison was already made by Fick and Prellwitz in the nineteenth century — Irish *derna* was compared with Greek δῶρον in its palm-breadth sense and both referred to PIE **der-*. The philological intuition was sound; what was lacking was the formal account that the bare-element architecture now supplies.

4.4 Old Irish *derna* and Continental Celtic *DERNV*

Old Irish *derna*, Gaelic *deàrna* "palm of the hand" is reinstated in v26 as the direct Celtic surface witness for the unmarked stem **de-r-nā-*. Earlier versions of this paper downgraded it following Matasóvič's decision not to reconstruct it as Proto-Celtic; that downgrading was specific to a **deuH-r-* reconstruction that no longer forms part of the framework. Under the bare-element architecture, *derna* is the simple unmarked stem **de-r-na-* — morphologically transparent and requiring no special commitment beyond the framework's core claims.

Continental Celtic attestation reinforces this. Stifter (2022: §10), in his etymological analysis of the Bronze of Novallas (Z.02.01), proposes that the fragmentary segment **DERNV* in line 8 is comparable with Old Irish feminine n-stem *dernu*, genitive *dernann* "palm of the hand," and sets up a Proto-Celtic preform **dernaŋon-*. The reading is offered as an alternative to the editorial reading by Beltrán Lloris et al. (2021), and Stifter explicitly notes that the relationship between Old Irish *dernu* "palm" and *dorn* "fist" (and the broader **durno-* family) "cannot be a simple case of ablaut."

That observation is, under the framework's morphological analysis, exactly what is predicted. The relationship between *derna* and **durno-* is not ablauting but morphological: unmarked stem **de-r-na-* (the open hand, the default) versus compound stem **der-u-no-* (the closed hand, the marked form). Stifter's caveat, which v25 of this paper had to absorb as a constraint, is under the present architecture a confirmation.

5. The Baltic evidence: Latvian *dūre*

Latvian *dūre*, *dūris* "fist" enters the cognate set as a third Indo-European witness. Schrijver (1995: 58) lists it explicitly among the "possible cognates" of Welsh *dwrn* and analyzes the long *ū* as reflecting earlier **uH*. Derksen (2015) takes an agnostic stance on the Baltic **dur-* family, provisionally separating it from PIE **der-/*dir-*; notably, his dictionary has no entry for *dūre* "fist" at all. The form is isolated within Baltic with no Lithuanian cognate.

Under the framework's architecture, three readings of Latvian *dūre* are open: (a) a parallel Baltic formation on the same body-part noun **de-r-*, with a Baltic-internal morphological development producing the long *ū*; (b) a separate Baltic body-part word with surface resemblance by coincidence or *Wanderwort*; (c) a connection to the Lithuanian *durti* "stab, prick" family rather than to a body-part etymon. The framework registers option (a) as possible without making it load-bearing. The Greek and Celtic evidence is sufficient to anchor the lexical claim; the Latvian form's status is left as an open question for Baltic-internal philology.

6. The Italic evidence: Latin *dōnum* and *dare*

Latin *dōnum* "gift" parallels Greek *δῶρον* as the Italic reflex of the gift-noun **deh₃-no-* — the n-stem alternant of the same r/n-heteroclite whose r-stem alternant yields Greek *δῶρον*. Under the bare-element architecture, both gift-words are derived from the verbal stem **deh₃-* (= **de-* + verbal extension **-h₃-*) with further nominalizing extensions. The body-part senses preserved in Greek and Celtic have no Italic surface witness, but Italic preserves both the verbal stem (Latin *dare*) and the gift-noun (*dōnum*).

The short *a* of Latin *dare* reflects the regular treatment of zero-grade **dh₃-* with laryngeal vocalization. The long *ō* of *dō* "I give" reflects the regular o-grade **doh₃-* with laryngeal coloring and lengthening. These are standard derivations requiring no framework-specific commitment beyond the bare-element decomposition of Paper 3.

7. The Anatolian convergence

The framework receives independent semantic support from Anatolian, on grounds developed by Norbruis (2021: 243–249) without reference to the present project. Norbruis argues, on Anatolian-internal morphological evidence, that the original meaning of the root reconstructed as **deh₃-* was "to take," not "to give": the give-meaning is a post-Anatolian innovation, with Anatolian *dā-* "take" preserving the older sense. His semantic development runs: "to take" > (with Goal or Beneficiary constituent) "to bring, convey, deliver" > (when the constituent was animate) "to hand, to give." The Anatolian branch preserves only the original meaning because it split off before the development started.

Norbruis (2021: 245 n. 9) also reports that Boley (2007) previously proposed deriving **deh₃-* from "a directionally ambivalent particle reconstructed on the basis of Latin *dē* 'from' and Greek *-δε* 'to'" — the same Greek allative clitic that Paper 3 adduces for the bare **de-* element. Paper 4 develops this insight into a full account: Hittite *dā-* "to take" is a fossilized univerbation of bare **de-* with a postposed directional element **a* "toward the speaker," contracted by hiatus to long **ā* in pre-Anatolian, at a stage before the verbal extension **-h₃-* was added.

The Anatolian convergence does double work for the framework. It provides a published specialist's independent argument that the original meaning was something concrete and physical and the abstract "give" meaning is secondary — consistent with a body-part origin in which the hand that takes and the hand that gives are two perspectives on a single physical act. And it places Boley's (2007) directional-particle intuition in the Anatolianist literature, giving Paper 3's decomposition of bare **de-* a published precursor.

8. Phonological account

8.1 Greek δῶρον- via the standard **deh₃-ro-* derivation

Greek δῶρον "gift" derives from **deh₃-ro-* by the standard derivation: the o-grade form **doh₃-ro-* undergoes regular laryngeal coloring and compensatory lengthening to yield **dō-ro-* > δῶρον. The framework's contribution at the phonological level is not in this surface derivation, which is unchanged, but in the morphological-architectural reading of **deh₃-* itself: Paper 3 segments it as bare **de-* + verbal extension **-h₃-*. The laryngeal is real and present in the verbal stem; the framework does not propose to remove it. Its status as a separable derivational extension rather than a root-internal constituent is what the bare-element architecture establishes.

Hesychian δάρις, by contrast, derives from the locative-extended body-part noun **de-r-* with an *i*-stem suffix: **de-r-i-s* or its zero-grade variant **d-r-i-s* with regular Greek α-vocalism. No laryngeal needed, no *u* needed. The long ω of δῶρον and the α of δάρις are, on this analysis, the respective traces of the verbal extension **-h₃-* (which colors and lengthens the vowel) and its absence (the plain locative formation, without laryngeal). The "obscure" formal relation that Schrijver flagged is thus the family relation between two sibling derivations from **de-*, one through the verbal extension and one through the locative extension.

8.2 Celtic **durno-* via morphological **u* in the compound stem

The Celtic surface form **durno-* is the morphologically-marked compound-stem reflex of the body-part noun **de-r-*, with the *u* arising from regular Celtic compound-stem morphology operating on the compositional stem (§4.2). The unmarked stem **de-r-* surfaces directly as Old Irish *derna* (with feminine singulative **-nā-*). The compound stem **der-u-* + singulative **-no-* yields **deruno-*

> **durno-* by regular vowel reduction at the morpheme boundary.

McCone (1996) and Matasóvič (2009, s.v. **g^wan-o-*) provide the methodological parallel. PIE **g^wh₂η-* "to kill" — a zero grade with syllabic nasal — yields Old Irish *gonaid* and Welsh *gwanu*. McCone states: "The o-vocalism in OIr. *gonaid* is the result of the regular rounding after the initial **g^w-*." The *o* in *gonaid* is not a PIE root vowel; it is a secondary product of the labiovelar environment. The same methodological point applies here: the *u* of Celtic **durno-* is the regular product of Celtic morphological processes operating on the inherited stem, not a direct witness to a PIE root *u*.

9. Engaging the literature

9.1 Matasóvič's substrate classification

Matasóvič's *Etymological Dictionary of Proto-Celtic* (2009, s.v. **durno-*) states: "The IE etymology of these Celtic words in IEW (203) cannot be accepted (Alb. *dorë* 'hand' is from PIE **ǵ^hesr-* 'hand,' Gr. *kheir*, etc.). Probably a loanword from a non-IE language." Matasóvič rejects the IEW etymology on a single ground: the removal of Albanian *dorë* from the cluster (Demiraj 1990 derived it from **ǵ^hesr-*). With Albanian removed, he finds no satisfactory IE etymology and falls back to substrate classification.

The framework offers exactly the alternative IE etymology that was missing. The reconstruction of the body-part noun as compositional **de-r-* does not depend on Albanian. Its evidential base is: Greek (δάρις as the primary body-part form, family-related δῶρον and δάρυος), Celtic (**durno-* and the wrist/hilt compounds, Old Irish *derna*, Continental Celtic **DERNVI*), Italic (gift-noun *dōnum*, verbal *dare*), and Anatolian (Paper 4). All of this evidence survives after the removal of Albanian. Matasóvič's substrate classification was a placeholder for a missing etymology; the framework supplies the etymology under the bare-element architecture.

9.2 Mayrhofer's rejection of an independent **der-*

Mayrhofer (1992: 703), in his entry on Sanskrit *ādara-* "esteem, respect," explicitly rejects an independent IE root **der ~ *derk* attributed to Grassmann (1875: 625). His verdict: the data "does not establish an independent IE root form" and the relevant senses are better derived from the standard **DARⁱ* "to break open" + the prefix *ā-*.

The hypothesis Mayrhofer rejects and the hypothesis proposed here are not the same. Mayrhofer rejects Grassmann's **der-* with semantics "esteem, respect, regard, pay attention to" — a different semantic field, a different evidence base (Sanskrit *ādara-* and related forms), and a different morphological structure (bare root, no decomposition). The framework proposes **de-r-* "hand" — a compositional

body-part noun on a different evidence base: Greek δῆρις, Celtic **durno-/*derna-*, Latvian *dūre*. Mayrhofer did not consider this body-part hypothesis when he rejected Grassmann's. The two rejections target different proposals.

9.3 Beekes's objection: surface *u* as PIE *u*

Beekes (*EDG* s.v. δῶρον, second lemma) states: "Further Celt. words, like OIr. *dorn*, as well as Latv. *dure*, *duris* 'fist', have been connected, but both branches point to PIE **u*, which excludes any relationship with δῶρον." Beekes treats the Celtic and Baltic *u* as witnesses to PIE root vocalism and concludes that the vocalism is incompatible with the gift-verb connection.

The framework splits the difference with Beekes. It agrees that the Celtic *u* is original to Celtic branches — not a secondary product of Brythonic raising — but disagrees on the conclusion. Beekes treats original-to-Celtic *u* as proof of an incompatible PIE root. The framework treats it as evidence about the Celtic morphological system: the *u* is morphological-stage Celtic, introduced by Celtic-internal compound-stem morphology operating on the inherited compositional stem **de-r-*. Once the Celtic *u* is recognized as Celtic-stage morphological rather than inherited PIE-stage lexical, Beekes's objection is dissolved: the surface *u* is no longer a witness to PIE root vocalism, so its presence is not a problem for the cognate connection.

10. Costs, alternatives, and what we do not claim

The reconstruction of the body-part noun as compositional **de-r-* is non-standard. It does not propose a new PIE root — the bare **de-* and the locative **-r-* are independently motivated in Paper 3 — but it does propose a body-part noun that has not been in the standard handbooks. Reviewers committed to atomistic root reconstruction will resist the compositional analysis on principle.

The Celtic *u* mechanism is left underspecified: three live candidates are registered (§4.2) without commitment to a specific Celtic-internal process. This is honest underspecification — the framework can identify the morphological-stage character of the *u* without specifying which process introduced it — but a future paper in Celtic-internal morphology should sharpen the analysis.

The body-part "palm-breadth" sense of Greek δῶρον is treated as secondary (preserved by family-association rather than direct inheritance from a body-part root). Earlier versions of this paper treated δῶρον's dual attestation as the argument's emotional center. That rhetorical role is now carried by the Hesychian δῆρις, Old Irish *derna*, and Continental Celtic **DERNV*[-]; δῶρον is a family-associated witness, not the primary anchor. The body-part evidence is more diffuse under v26 but more defensible architecturally.

Latvian *dūre* is registered as a possible cognate but not load-bearing. The long *ū* needs a separate Baltic-internal account that this paper does not commit to.

We do not claim **deh₃-* is wrong as the verbal pre-form for the give-verb. We treat it as correct under standard laryngeal phonology; Paper 3's decomposition of **deh₃-* as **de-* + **-h₃-* is a morphological-architectural claim, not a phonological one, and it does not modify the standard surface derivations of δίδωμι, *dare*, *dā-*, or *dádāti*. We do not claim that all body-part "hand" words across Indo-European descend from **de-r-*. PIE had multiple hand-words; **ǵz^hesr-* (Greek χεῖρ, Albanian *dorë*, Hittite *keššar*) is a separate hand-word with its own distribution and morphology.

11. Probability assessment

We continue to assess the framework's probability honestly. The probability for Paper 1 alone — that the comparative cluster Greek δάρις / Celtic **durno-/*derna-* / Latvian *dūre* reflects a body-part noun **de-r-* "hand" — is assessed at approximately **25-35%**. This is an upward revision from v23's 15-20%, reflecting three changes: the architectural consistency now achieved across the four papers (no tension between Paper 1's reconstruction and Papers 2-4); the parsimony of the bare-element account (no novel PIE root **deuH-r-* required); and the engagement with Stifter's (2022) caveat as confirmation rather than constraint. The offsetting costs are the demotion of δῶρον from primary body-part witness to family-associated gift-noun, and the underspecified Celtic *u* mechanism.

The marginal probability for the whole four-paper framework — unconditioned, integrating over the probability that each successive paper is also correct — is approximately: 25-35% (Paper 1) × 35-45% (Paper 2 conditional on Paper 1) × 32-42% (Paper 3 conditional on Papers 1-2) × 38-48% (Paper 4 conditional on Papers 1-3) ≈ 1.1-2.6%. This low marginal figure represents the cumulative ambition of the framework accurately. We continue to regard the central claim as more likely wrong than right, and to regard it as interesting enough, and the evidential structure dense enough, that publication for examination by other scholars is warranted. These two judgments are compatible.

12. Conclusion

The Greek-Celtic-Baltic hand cluster has stood since the nineteenth century as a comparative set whose formal derivation was unresolved. Pokorny (IEW 203) listed it; Schrijver (1995: 58) called the formal relation "obscure"; Matasóvič (2009) classified the Celtic word as substrate after concluding that no IE etymology was satisfactory. The present paper offers an Indo-European etymology under the bare-element architecture of the broader four-paper project: the body-part noun is the compositional **de-r-* (= bare **de-* of Paper 3 + locative **-r-* of Paper 3, supported by Lipp 2009, Bauhaus 2019, Kahl 2024).

The body-part noun proper is preserved in Hesychian δάρις (Greek zero-grade *i*-stem body-part form), Old Irish *derna* (unmarked feminine singular **de-r-nā-*),

and Continental Celtic **DERNV* (Bronze of Novallas, Stifter 2022). The Celtic compound stem **durn-* — attested in Insular Celtic **durno-* "fist," in the wrist and sword-hilt compounds confirming the "hand" meaning, and in Continental Gaulish *Durno-* — arises from regular Celtic morphology operating on the compositional stem; the *u* is morphological-stage Celtic, not inherited PIE root vocalism. Greek δῶρον is the gift-noun **deh₃-ro-* from the verbal stem; the body-part "palm-breadth" sense is preserved by family-association with the sibling **de-r-*.

The framework's contribution under v26 is architectural unification. Papers 1, 2, 3, and 4 now reconstruct five parallel derivatives of a single bare hand-element **de-*: the directional particle (**de/do*), the body-part noun (**de-r-*), the verbal stem (**deh₃-*), the contrastive suffix (**-tero-*), and the Anatolian take-verb (**de + a > Hittite dā-*). The framework requires no novel PIE root and no novel laryngeal phonology. It requires only the bare-element architecture itself, which Papers 2, 3, and 4 have independently motivated. The IE etymology Matasóvič noted was missing is now on the table.

13. Authorship and contributions

This is the twenty-sixth version of Paper 1 in the *der-* project. It builds on Papers 2, 3, and 4 (Chavis & Claude 2026b, 2026c, 2026d) and continues the same co-authorship arrangement.

Chavis's contributions. The central hypothesis originates with Chavis. The identification of the two attested senses of Greek δῶρον — gift and palm-breadth — as evidence of a single etymological family was Chavis's core insight. Chavis directed the research agenda throughout, made all fundamental decisions about scope and framing, recognized the inconsistency between earlier versions' **deuH-r-* reconstruction and Paper 3's bare-element segmentation, and directed the v26 architectural revision that resolves it. The reframing of Stifter's (2022) "cannot be a simple case of ablaut" observation from constraint to confirmation is Chavis's judgment.

Claude's contributions. Claude's contributions are primarily in source synthesis, exposition, and analytical development across versions. Claude located and read the relevant published literature throughout the project, identified the Kloekhorst (2013) and Norbruis (2021) literature, surfaced Stifter (2022) and Bauhaus (2019) as new sources, integrated the architectural alignment across the four papers, and drafted the prose from Chavis's structural and substantive direction. The probability estimate in §11 is Claude's own. Claude is an AI system without persistent memory between sessions; the intellectual continuity of the project is supplied by Chavis through systematic transcript management. The collaboration is genuine. The argument was built together. Its strengths and weaknesses belong to both authors, with the responsibility resting with Chavis.

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